

# BANK ACCOUNT MANAGEMENT

## Java Programming Assignment

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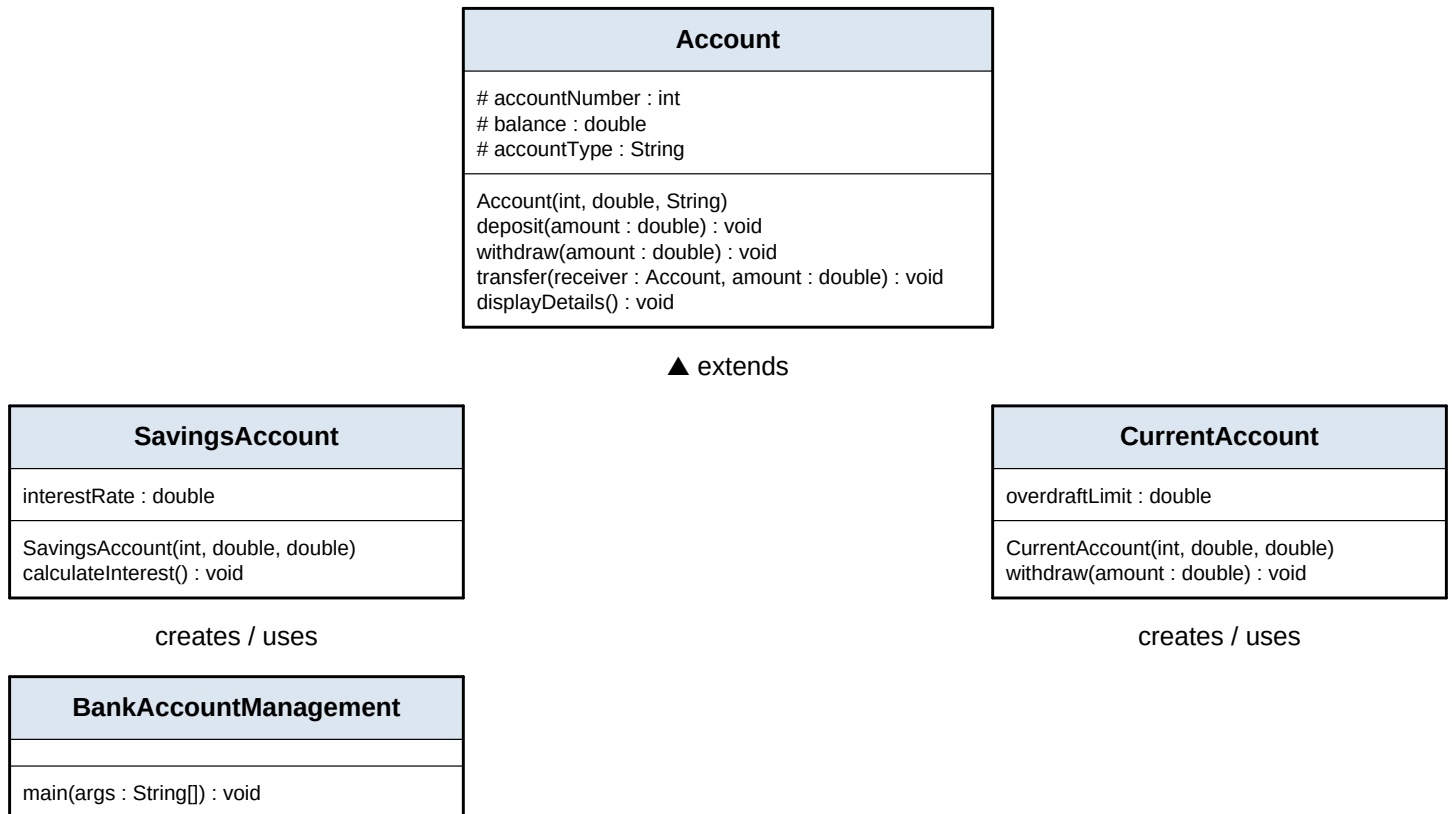
# 1. ALGORITHM

Steps for executing the Bank Account Management program:

| Step | Algorithm                                                                                |
|------|------------------------------------------------------------------------------------------|
| 1    | Start the program.                                                                       |
| 2    | Create SavingsAccount s1 with account number 101, balance 10000 and interest rate 5%.    |
| 3    | Create CurrentAccount c1 with account number 102, balance 5000 and overdraft limit 3000. |
| 4    | Display Savings Account details.                                                         |
| 5    | Deposit 2000 into the Savings Account.                                                   |
| 6    | Withdraw 3000 from the Savings Account.                                                  |
| 7    | Calculate and display interest.                                                          |
| 8    | Display the Savings Account balance.                                                     |
| 9    | Display Current Account details.                                                         |
| 10   | Deposit 2000 into the Current Account.                                                   |
| 11   | Withdraw 9000 using the overdraft facility.                                              |
| 12   | Display the Current Account balance.                                                     |
| 13   | Transfer 2000 from Savings Account to Current Account.                                   |
| 14   | Display both account balances after transfer.                                            |
| 15   | Stop the program.                                                                        |

## 2. UML CLASS DIAGRAM

SavingsAccount and CurrentAccount inherit from Account. BankAccountManagement creates and uses both objects.



**Note:** # denotes protected members. CurrentAccount overrides withdraw().

### 3. JAVA SOURCE CODE

File name: **BankAccountManagement.java**

```
class Account {
    protected int accountNumber;
    protected double balance;
    protected String accountType;

    Account(int accountNumber, double balance, String accountType) {
        this.accountNumber = accountNumber;
        this.balance = balance;
        this.accountType = accountType;
    }

    void deposit(double amount) {
        balance = balance + amount;
        System.out.println("Deposited: " + amount);
    }

    void withdraw(double amount) {
        if (amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Insufficient balance");
        }
    }

    void transfer(Account receiver, double amount) {
        if (amount <= balance) {
            balance = balance - amount;
            receiver.balance = receiver.balance + amount;
            System.out.println("Transfer successful: " + amount);
        } else {
            System.out.println("Insufficient balance for transfer");
        }
    }

    void displayDetails() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Account Type: " + accountType);
        System.out.println("Balance: " + balance);
    }
}

class SavingsAccount extends Account {
    double interestRate;
}
```

### 3. JAVA SOURCE CODE (CONTINUED)

```
SavingsAccount(int accountNumber, double balance, double interestRate) {
    super(accountNumber, balance, "Savings Account");
    this.interestRate = interestRate;
}

void calculateInterest() {
    double interest = balance * interestRate / 100;
    System.out.println("Interest: " + interest);
}
}

class CurrentAccount extends Account {
    double overdraftLimit;

    CurrentAccount(int accountNumber, double balance, double overdraftLimit) {
        super(accountNumber, balance, "Current Account");
        this.overdraftLimit = overdraftLimit;
    }

    @Override
    void withdraw(double amount) {
        if (amount <= balance + overdraftLimit) {
            balance = balance - amount;
            System.out.println("Withdrawn: " + amount);
        } else {
            System.out.println("Overdraft limit exceeded");
        }
    }
}

public class BankAccountManagement {
    public static void main(String[] args) {

        SavingsAccount s1 =
            new SavingsAccount(101, 10000, 5);

        CurrentAccount c1 =
            new CurrentAccount(102, 5000, 3000);

        System.out.println(" SAVINGS ACCOUNT ");

        s1.displayDetails();
    }
}
```

### 3. JAVA SOURCE CODE (CONTINUED)

```
s1.deposit(2000);
s1.withdraw(3000);
s1.calculateInterest();
System.out.println("Balance after transactions: " + s1.balance);

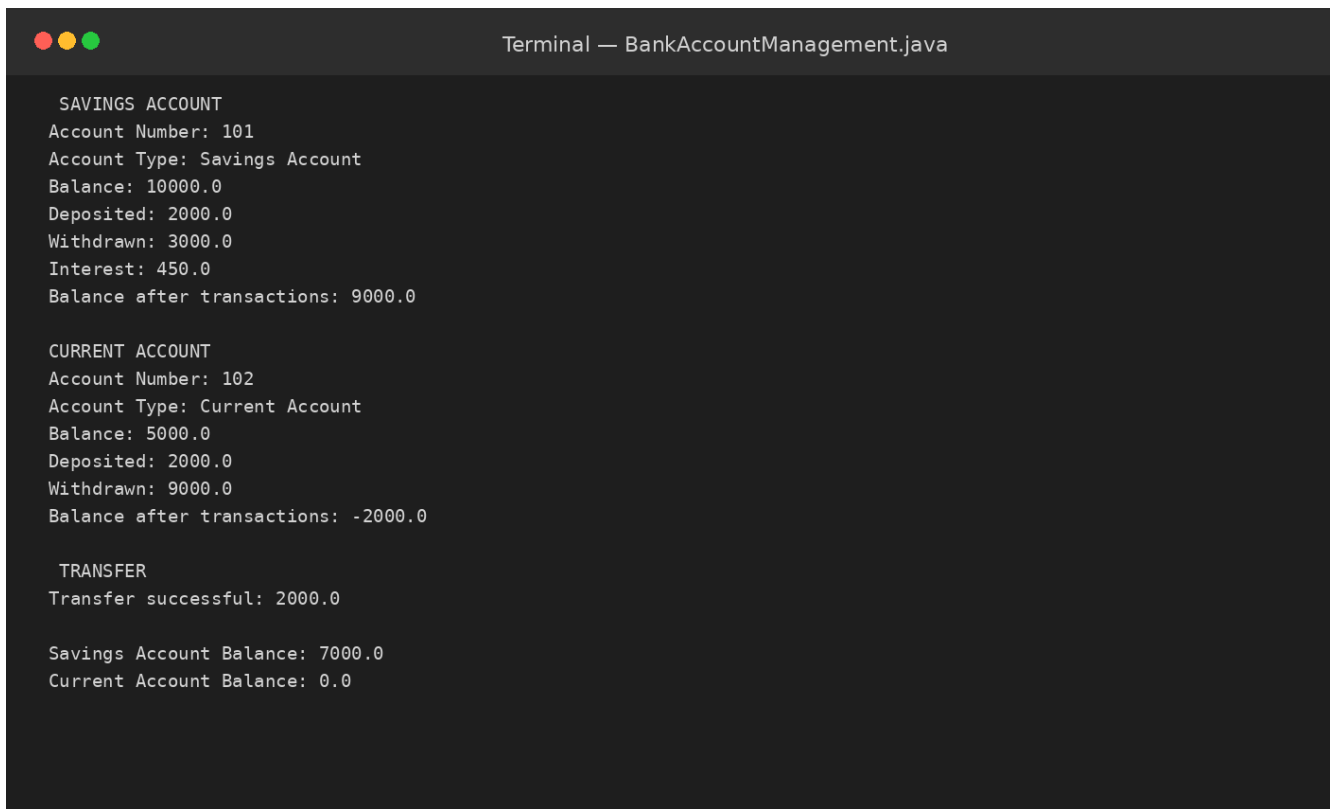
System.out.println("\nCURRENT ACCOUNT ");
c1.displayDetails();
c1.deposit(2000);
c1.withdraw(9000);
System.out.println("Balance after transactions: " + c1.balance);

System.out.println("\n TRANSFER ");
s1.transfer(c1, 2000);

System.out.println("\nSavings Account Balance: " + s1.balance);
System.out.println("Current Account Balance: " + c1.balance);
    }
}
```

## 4. OUTPUT SCREEN – VS CODE STYLE

The following screen is formatted in the style of a VS Code integrated terminal.

A screenshot of a VS Code integrated terminal window. The title bar at the top shows three colored window control buttons (red, yellow, green) on the left and the text "Terminal — BankAccountManagement.java" on the right. The terminal area has a dark background with light gray text. The output is formatted with section headers in all caps and indented lines for details. The sections are: SAVINGS ACCOUNT, CURRENT ACCOUNT, TRANSFER, and final balance statements.

```
SAVINGS ACCOUNT
Account Number: 101
Account Type: Savings Account
Balance: 10000.0
Deposited: 2000.0
Withdrawn: 3000.0
Interest: 450.0
Balance after transactions: 9000.0

CURRENT ACCOUNT
Account Number: 102
Account Type: Current Account
Balance: 5000.0
Deposited: 2000.0
Withdrawn: 9000.0
Balance after transactions: -2000.0

TRANSFER
Transfer successful: 2000.0

Savings Account Balance: 7000.0
Current Account Balance: 0.0
```

**Final balances:** Savings Account = 7000.0; Current Account = 0.0.